Gamma ray astrophysics: Assignment II

Astroparticle physics 2026

Due date: 1st May, 2026

1. Assume you measured n_{on} counts in a region where you think a source might be present. If the expected number of background events, n_{bkg} is known, derive the Maximum Likelihood Estimate for the expected number of signal counts from the region. This is commonly known as *Cash* or *CStat* statistic.

   - For $n_{on} = 10$ and $n_{bkg} = 3$, calculate the excess, significance of detection and the 1 and 2-sigma confidence intervals for the excess. Plot the likelihood profiles.
   - For $n_{on} = 5$ and $n_{bkg} = 3$, calculate the excess, significance of detection and the $5 - \sigma$ upper limit on the excess. Plot the likelihood profiles.

2. Same as above, but instead of having a model predicted number of background events, you have measured n_{off} counts in a background control region where you assume no source is present. The off region is twice as large as the on region, so $\alpha = \alpha_{on}/\alpha_{off} = 1/2$. Likelihood Estimate for the expected number of signal counts from the region. This is commonly known as *WStat* statistic.

   - For $n_{on} = 10$ and $n_{off} = 6$, calculate the excess, significance of detection and the 1 and 2-sigma confidence intervals for the excess. Plot the likelihood profiles.
   - For $n_{on} = 5$ and $n_{off} = 6$, calculate the excess, significance of detection and the $5 - \sigma$ upper limit on the excess. Plot the likelihood profiles.
   - Comment on the differences between the two likelihoods.

3. You are given a 1D signal composed of multiple Gaussian pulses with different widths. Study how convolution with a top-hat (boxcar) kernel of varying radii affects the detectability of these pulses. Create jupyter notebook showing in detail:

   - Construct a 1D time series with sufficient resolution (decide binning etc yourself).

- Generate a signal consisting of multiple Gaussian pulses. Choose at least 4 pulses of different amplitudes $\sigma = [0.01, 0.1, 1, 10]$ and place them at different positions in time. You may keep amplitudes same.
- Define top hat kernels of different radii and convolve with the signal.
- How does the kernel radius compare to the Gaussian width? Interpret your results.
- Compute the Fourier transform of the signal and kernel. Interpret.

4. Redo the above by adding a Poission noise on the time series. You can interpret it then as the number of detected counts from a certain object. Add slowly varying background on top. This is very similar to what we observe in blazar flares, and can be generalised to a study of flare detections.

5. Generalise (3) to source detections. Use Gammapy to simulate a 2D gamma-ray sky with sources of different spatial extensions and analyze it using *ExcessMapEstimator* and *TSMapEstimator*. Simulate a point source (PSF limited) and sources of different Gaussian widths. Follow for simulation and to understand how to use the estimators.

   - For each radii, plot the TS and excess maps, and identify statistically significant signals.
   - Discuss your results
   - What is the difference in interpretation between the two estimators? Which one will you prefer for claiming source detections?